

Artificial Intelligence Basics

What is Artificial Intelligence?

Artificial Intelligence (AI) is the ability of machines to simulate human intelligence and perform tasks such as learning, reasoning, and problem-solving.

Why is AI compared to chess?

AI is compared to chess because it learns from mistakes and improves its performance over time, just like a player who gets better after each game.

What is Machine Learning?

Machine Learning is a subset of AI where machines learn from data and improve their performance without being explicitly programmed.

Types of Machine Learning

Supervised Learning

Supervised learning is a type of machine learning where models are trained using labeled data, meaning the input comes with correct answers.

Unsupervised Learning

Unsupervised learning involves training models on data without labels, allowing the system to find patterns on its own.

Reinforcement Learning

Reinforcement learning is a method where an agent learns by interacting with an environment and receiving rewards or penalties.

What is Deep Learning?

Deep learning is a subset of machine learning that uses neural networks with multiple layers to analyze complex data.

What is ELIZA?

ELIZA is one of the first chatbot computer programs, developed in the 1960s, that simulated conversation with users.

Applications of AI

AI is used in many fields such as healthcare for diagnosis, Netflix for recommendations, cybersecurity for threat detection, and voice assistants like Siri and Alexa.